

The Dominant Pole Approximation

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Higher-Order Transfer Function

The transfer function we will analyse is 5th order:

$$\begin{aligned} T(s) &= \frac{12614(s+5)(s+6-j8)(s+6+j8)}{5(s+7)(s+5-j9)(s+5+j9)(s+1-j4)(s+1+j4)} \\ &= \frac{12614s^3 + 214438s^2 + 2018240s + 6307000}{5s^5 + 95s^4 + 1135s^3 + 6915s^2 + 22375s + 63070} \end{aligned}$$

Dominant Pole Approximation

The simplified transfer function is:

$$\begin{aligned} T_{dp}(s) &= \frac{1700}{(s+1-j4)(s+1+j4)} \\ &= \frac{1700}{s^2 + 2s + 17} \end{aligned}$$